CrowdCounter
751 tensors total (969.9 MB)
121518770 params (33954354/121518770 trainable, 463.6 MB)

input_1
(1, 3, 256, 256), 768.0 KB
@input.x
identity_110_559
(1, 3, 256, 256), 768.0 KB
conv2d_1_1
(1, 128, 64, 64), 2.0 MB
in_channels=3, out_channels=128, kernel_size=(4, 4), stride=(4, 4)
params: weight [128, 3, 4, 4] · bias [128,]
@net.en_de.encoder.model.stem_0
@net.en_de.encoder.model.stem_1
LayerNorm2d
(1, 128, 64, 64), 2.0 MB
3 ops
256 params (all frozen)
@net.en_de.encoder.model.stages_0
ConvNeXtStage
(1, 128, 64, 64), 2.0 MB
46 ops
415488 params (all frozen)
@net.en_de.encoder.model.stages_1
ConvNeXtStage
(1, 256, 32, 32), 1.0 MB
49 ops
1748992 params (all frozen)
@net.en_de.encoder.model.stages_2
ConvNeXtStage
(1, 512, 16, 16), 512.0 KB
409 ops
57950208 params (all frozen)
@net.en_de.encoder.model.stages_3
ConvNeXtStage
(1, 1024, 8, 8), 256.0 KB
49 ops
27443200 params (all frozen)
@net.en_de.encoder.model
(FeatureListNet)
@net.en_de.encoder
(TimmUniversalEncoder)
conv2d_42_561
(1, 64, 8, 8), 16.0 KB
in_channels=1024, out_channels=64, kernel_size=(1, 1)
params: weight (64, 1024, 1, 1) · bias (64,)
@net.en_de.decoder.center.center_conv
conv2d_41_560
(1, 64, 8, 8), 16.0 KB
in_channels=1024, out_channels=64, kernel_size=(1, 1)
params: weight (64, 1024, 1, 1) · bias (64,)
@net.en_de.decoder.center.top_conv
flatten_2_564
(1, 64, 64), 16.0 KB
flatten_1_563
(1, 64, 64), 16.0 KB
transpose_1_565
(1, 64, 64), 16.0 KB
dim0=1, dim1=2
arg 1
arg 0
matmul_1_568
(1, 64, 64), 16.0 KB
conv2d_43_562
(1, 1024, 8, 8), 256.0 KB
in_channels=1024, out_channels=1024, kernel_size=(3, 3), padding=(1, 1)
params: weight (1024, 1024, 3, 3) · bias (1024,)
@net.en_de.decoder.center.bottom_conv
view_1_569
(1, 4096), 16.0 KB
softmax_1_570
(1, 4096), 16.0 KB
dim=1
@net.en_de.decoder.center.map_softmax
flatten_3_566
(1, 1024, 64), 256.0 KB
transpose_2_567
(1, 64, 1024), 256.0 KB
dim0=1, dim1=2
view_2_571
(1, 64, 64), 16.0 KB
arg 1
arg 0
matmul_2_572
(1, 64, 1024), 256.0 KB
reshape_37_573
(1, 1024, 8, 8), 256.0 KB
add_37_574
(1, 1024, 8, 8), 256.0 KB
conv2d_44_575
(1, 1024, 8, 8), 256.0 KB
in_channels=1024, out_channels=1024, kernel_size=(3, 3), padding=(1, 1)
params: weight (1024, 1024, 3, 3) · bias (1024,)
@net.en_de.decoder.center.out_conv
@net.en_de.decoder.center
(PABBlock)
@net.en_de.decoder.blocks.0.hl_conv
Sequential
(1, 512, 8, 8), 128.0 KB
6 ops + 8 buffers
9964544 params (all trainable)
interpolate_1_582
(1, 512, 16, 16), 512.0 KB
scale_factor=2.0, mode=nearest
@net.en_de.decoder.blocks.0.SE_hl
Sequential
(1, 512, 1, 1), 2.0 KB
5 ops
33312 params (all trainable)
@net.en_de.decoder.blocks.0.SE_ll
Sequential
(1, 512, 1, 1), 2.0 KB
5 ops
33312 params (all trainable)
add_38_593
(1, 512, 1, 1), 2.0 KB
mul_37_594
(1, 512, 16, 16), 512.0 KB
cat_1_595
(1, 1024, 16, 16), 1.0 MB
dim=1
@net.en_de.decoder.blocks.0.conv1
Conv2dReLU
(1, 256, 16, 16), 256.0 KB
3 ops + 4 buffers
2359808 params (all trainable)
@net.en_de.decoder.blocks.0.conv2
Conv2dReLU
(1, 256, 16, 16), 256.0 KB
3 ops + 4 buffers
590336 params (all trainable)
@net.en_de.decoder.blocks.0
(MFABBlock)
@net.en_de.decoder.blocks.1.hl_conv
Sequential
(1, 256, 16, 16), 256.0 KB
6 ops + 8 buffers
656384 params (all trainable)
interpolate_2_608
(1, 256, 32, 32), 1.0 MB
scale_factor=2.0, mode=nearest
@net.en_de.decoder.blocks.1.SE_hl
Sequential
(1, 256, 1, 1), 1.0 KB
5 ops
8464 params (all trainable)
@net.en_de.decoder.blocks.1.SE_ll
Sequential
(1, 256, 1, 1), 1.0 KB
5 ops
8464 params (all trainable)
add_39_619
(1, 256, 1, 1), 1.0 KB
mul_38_620
(1, 256, 32, 32), 1.0 MB
cat_2_621
(1, 512, 32, 32), 2.0 MB
dim=1
@net.en_de.decoder.blocks.1.conv1
Conv2dReLU
(1, 128, 32, 32), 512.0 KB
3 ops + 4 buffers
590080 params (all trainable)
@net.en_de.decoder.blocks.1.conv2
Conv2dReLU
(1, 128, 32, 32), 512.0 KB
3 ops + 4 buffers
147712 params (all trainable)
@net.en_de.decoder.blocks.1
(MFABBlock)
@net.en_de.decoder.blocks.2.hl_conv
Sequential
(1, 128, 32, 32), 512.0 KB
6 ops + 8 buffers
164352 params (all trainable)
interpolate_3_634
(1, 128, 64, 64), 2.0 MB
scale_factor=2.0, mode=nearest
@net.en_de.decoder.blocks.2.SE_hl
Sequential
(1, 128, 1, 1), 512 B
5 ops
2184 params (all trainable)
@net.en_de.decoder.blocks.2.SE_ll
Sequential
(1, 128, 1, 1), 512 B
5 ops
2184 params (all trainable)
add_40_645
(1, 128, 1, 1), 512 B
mul_39_646
(1, 128, 64, 64), 2.0 MB
cat_3_647
(1, 256, 64, 64), 4.0 MB
dim=1
@net.en_de.decoder.blocks.2.conv1
Conv2dReLU
(1, 64, 64, 64), 1.0 MB
3 ops + 4 buffers
147584 params (all trainable)
@net.en_de.decoder.blocks.2.conv2
Conv2dReLU
(1, 64, 64, 64), 1.0 MB
3 ops + 4 buffers
36992 params (all trainable)
@net.en_de.decoder.blocks.2
(MFABBlock)
empty_1_558
(1, 0, 128, 128), 0 B
interpolate_4_654
(1, 64, 128, 128), 4.0 MB
scale_factor=2.0, mode=nearest
cat_4_655
(1, 64, 128, 128), 4.0 MB
dim=1
@net.en_de.decoder.blocks.3.conv1
Conv2dReLU
(1, 32, 128, 128), 2.0 MB
3 ops + 4 buffers
18496 params (all trainable)
@net.en_de.decoder.blocks.3.conv2
Conv2dReLU
(1, 32, 128, 128), 2.0 MB
3 ops + 4 buffers
9280 params (all trainable)
@net.en_de.decoder.blocks.3
(DecoderBlock)
interpolate_5_662
(1, 32, 256, 256), 8.0 MB
scale_factor=2.0, mode=nearest
@net.en_de.decoder.blocks.4.conv1
Conv2dReLU
(1, 16, 256, 256), 4.0 MB
3 ops + 4 buffers
4640 params (all trainable)
@net.en_de.decoder.blocks.4.conv2
Conv2dReLU
(1, 16, 256, 256), 4.0 MB
3 ops + 4 buffers
2336 params (all trainable)
@net.en_de.decoder.blocks.4
(DecoderBlock)
@net.en_de.decoder
(MAnetDecoder)
conv2d_73_669
(1, 128, 256, 256), 32.0 MB
in_channels=16, out_channels=128, kernel_size=(3, 3), padding=(1, 1)
params: weight (128, 16, 3, 3) · bias (128,)
@net.en_de.segmentation_head.0
identity_111_670
(1, 128, 256, 256), 32.0 MB
@net.en_de.segmentation_head.1
identity_112_671
(1, 128, 256, 256), 32.0 MB
@net.en_de.segmentation_head.2.activation
identity_113_672
(1, 128, 256, 256), 32.0 MB
@net.en_de.segmentation_head.2
@net.en_de.segmentation_head
(SegmentationHead)
@net.en_de
(MAnet)
@net
(EncoderDecoder)
conv2d_74_673
(1, 64, 256, 256), 16.0 MB
in_channels=128, out_channels=64, kernel_size=(3, 3), padding=(1, 1)
params: weight (64, 128, 3, 3) · bias (64,)
@attention_head.0
relu_23_674
(1, 64, 256, 256), 16.0 MB
@attention_head.1
featuredropout_1_675
(1, 64, 256, 256), 16.0 MB
@attention_head.2
conv2d_75_676
(1, 1, 256, 256), 256.0 KB
in_channels=64, out_channels=1, kernel_size=(1, 1)
params: weight (1, 64, 1, 1) · bias (1,)
@attention_head.3
@attention_head
(Sequential)
conv2d_76_678
(1, 64, 256, 256), 16.0 MB
in_channels=128, out_channels=64, kernel_size=(3, 3), padding=(1, 1)
params: weight (64, 128, 3, 3) · bias (64,)
@density_features.0
relu_24_679
(1, 64, 256, 256), 16.0 MB
@density_features.1
featuredropout_2_680
(1, 64, 256, 256), 16.0 MB
@density_features.2
@density_features
(Sequential)
sigmoid_7_677
(1, 1, 256, 256), 256.0 KB
mul_40_681
(1, 64, 256, 256), 16.0 MB
add_41_682
(1, 64, 256, 256), 16.0 MB
conv2d_77_683
(1, 1, 256, 256), 256.0 KB
in_channels=64, out_channels=1, kernel_size=(1, 1)
params: weight (1, 64, 1, 1) · bias (1,)
@final_density_conv.0
relu_25_684
(1, 1, 256, 256), 256.0 KB
@final_density_conv.1
@final_density_conv
(Sequential)
truediv_1_685
(1, 1, 256, 256), 256.0 KB
output_1
(1, 1, 256, 256), 256.0 KB
@output